

Sara Sapre

B.Tech CSE Student at SRMIST

sarasapre57@gmail.com | 7796524234 | Mumbai, Maharashtra

linkedin.com/in/sara-sapre | github.com/laserFocusDev



SUMMARY

Second-year CSE undergraduate at SRMIST (CGPA: 9.84/10) with hands-on experience in deep learning, NLP, and computer vision. Seeking a research internship in ML/Deep Learning to contribute to generative models.

EDUCATION

2024 – 2028 Kattankulathur, Chennai	B.Tech CSE with specialization in Cyber Security <i>SRMIST- SRM Institute of Science and Technology</i> - current CGPA: 9.83 / 10.00 - Class Representative since 1st year, acting as liaison between faculty and students
2022 – 2024 Airoli, Maharashtra	Class XII <i>Arya Gurukul International College of Science and Commerce</i> 80.33% - Maharashtra State Board of Secondary and Higher Secondary Education
2021 – 2022 Pune, Maharashtra	Class X <i>Sanskriti School, Pune</i> 97.4% - Central Board of Secondary Education

PROJECTS

MF Portfolio X-Ray — Multi-Agent Financial Audit Engine

- Architected a 7-agent LangGraph DAG pipeline (Parser → XIRR → Overlap → Expense → Benchmark → Rebalance → Synthesis) achieving end-to-end portfolio analysis in 8.4 seconds; integrated Gemini 1.5 Flash for natural language synthesis of quantitative audit results
- Implemented Newton-Raphson method (scipy.optimize) for XIRR computation with bisection search fallback for convergence failures; built multi-strategy PDF parser handling 14+ CAMS/KFintech statement formats using positional coordinates and regex anchor-matching
- Engineered vectorized stock overlap detection across 200+ securities using pandas matrix operations (<50ms); developed real-time SSE streaming interface eliminating blocking wait states during agent execution

Pre-Delinquency Prediction Engine (Group Project)

- Built end-to-end ML pipeline on synthetic credit risk dataset (5,000 customers, aggregated from ~160M simulated behavioral log rows); engineered time-windowed features with strict prediction gap to prevent data leakage
- Trained LightGBM classifier achieving ROC-AUC 0.74 vs. 0.65 logistic regression baseline; detected and resolved an AUC=1.0 leakage bug during validation, recovering realistic model generalization

- Applied SHAP explainability to identify top predictive features and validate model behavior; managed structured log storage and querying via SQLite

NLP-Based News Classification using BERT

- Fine-tuned distilbert-base-uncased on ISOT Fake News dataset (5,000 articles, binary classification) with custom PyTorch Dataset class and WordPiece tokenization (max_length=128)
- Achieved 98.8% validation accuracy (F1: 0.99) vs. TF-IDF + Logistic Regression baseline of 94.2% (F1: 0.94) on identical 90/10 split; improvement most pronounced on context-dependent and opinion-heavy samples
- Conducted error analysis on misclassified samples and deployed end-to-end inference pipeline for real-time prediction on unseen articles

Image Classification using CNNs (MNIST & CIFAR-10)

- Implemented MNIST pipeline from scratch in PyTorch including raw IDX binary file parsing (without dataset libraries); trained 3-layer MLP (784→128→64→10) achieving 97% test accuracy with cross-entropy loss ≈ 0.09
- Designed custom 4-block CNN for CIFAR-10 (60K images, 10 classes) with BatchNorm and Dropout; improved test accuracy from 70% (2-layer baseline) to 81.9% through systematic architectural ablation
- Google colab link of the project: <https://colab.research.google.com/drive/16yB1-3PnLYpwxJC0TtRZOLGAYI11Gik?usp=sharing>

SKILLS

Programming Languages

Python
C/C++
Java

Frameworks

PyTorch, TensorFlow, HuggingFace Transformers, LangChain, LangGraph, FastAPI, scikit-learn, OpenCV

Machine Learning

Fine-tuning, SHAP Explainability, Multi-Agent Systems, Computer Vision, NLP, Time-Series Modeling

Tools

Git, SQLite, Pandas, NumPy, SciPy, Jupyter, Google Colab

ORGANISATIONS

06/2026 – 08/2026
Pune, Maharashtra

Arvillium AI Solutions

AI Engineering - Intern

AI engineering research intern on the finance-related machine learning project. Produced a written research paper and trained models and experimented on real data.

08/2025 – 11/2025

Robocon Club (Robotics Club)

Programming Team Member

Built a vision-based targeting system using OpenCV to detect targets and guide a 3-wheel drive robot. Converted visual input into movement for accurate alignment, and handled real-time debugging with hardware constraints.

02/2026 – Present

Liftoff Club

ML Trainee (Cohort 8)

Reading ML research papers and implementing models; currently studying attention mechanisms and transformer architectures

08/2024 – 04/2025

Directorate of Student Affairs- Creative Arts

Member

Contributed to creative design work for college events like Milan, helping with visuals and artistic elements for fest activities.

 AWARDS

06/01/2026

Performance Based Scholarship

SRM Institute of Science and Technology

Granted a fee waiver for Year II for having the top CGPA in the department.

 CERTIFICATES

AIML Workshop at IIT Madras (attended AI/ML Workshop at IIT Madras)

Malware Analysis workshop at SRMIST (attended Malware Analysis workshop at SRMIST)

Introduction to Cybersecurity (Certificate for completing "Introduction to Cybersecurity" by Cisco Networking Academy)

Deloitte Cyber Job Simulation (Certificate for completing a cyber job simulation by Deloitte)